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Homework 2: Neural Networks driven by Physics and convolutional neural networks for RGB Demosaicking

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21 de agosto de 2026

The assignment report is due on 2 September, 2026. In it, you must solve the exercises presented in the guide in report format, and attach the files you generate, such as .py, ipynb, .mat, .m, .fig, etc. The report should briefly describe the experiments performed, including the following sections: summary, description of experiments, and conclusions. It is suggested that you prepare the report in LaTeX to facilitate writing equations. Send a compressed folder with the files to the email address: **nelson.diazd@usm.cl**, with the subject line: **Homework2_FotografíaComputacional** and the compressed folder's name **homework2_nameSurname.zip**, where Name is your first name and Surname is your last name. Caution: any attempt at copying or fraud will invalidate the lab and result in a zero on the lab grade.

1. Learning Goal

To reconstruct from real-mosaic data capture with an electronic device, for instances, a smartphone, a tablet or laptop using physics informed neural networks and convolutional neural networks.

2. Introduction

The Bayer filter can be used on both charge-coupled device (CCD) and complementary Metal-Oxide-Semiconductor (CMOS) image sensor, when a measurement is captured and saved in digital negative format (DNG). The Bayer filter is able to subsampled red, green and blue of the visible spectrum 1. The task of demosaicking or RGB interpolation could be performed using traditional methods, *i.e.*, bilinear interpolation or using novel implicit neural representation 2 and convolutional neural networks (CNN). Implementations of both networks can be found here CNN and Implicit representation.

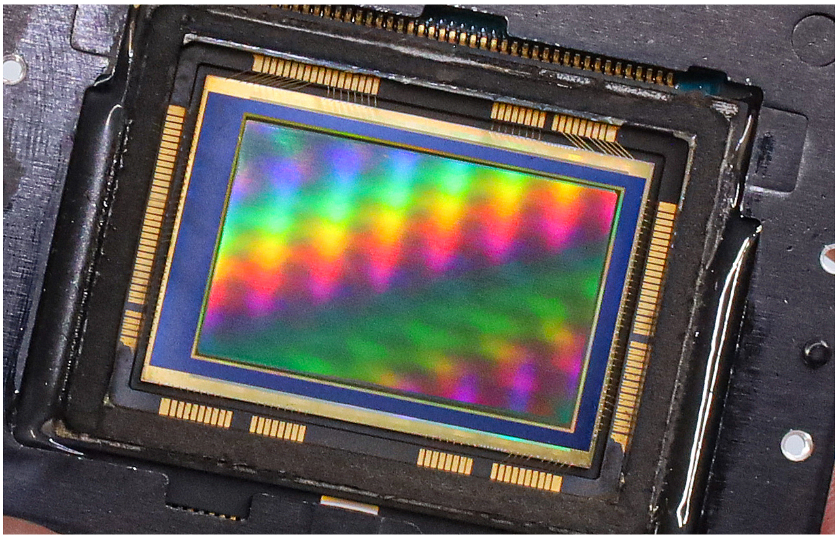

Figura 1: The Bayer filter pattern is placed over the pixels of a single sensor to capture red, green and blue light.

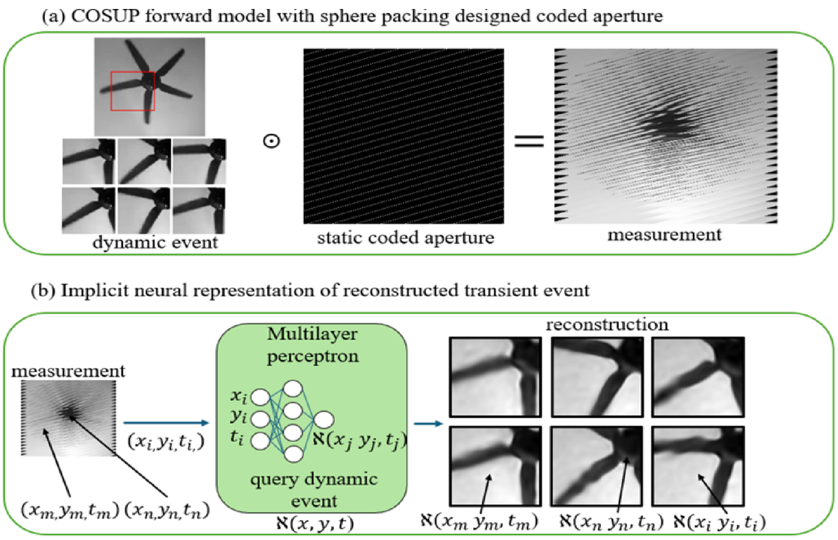

Figura 2: Physics informed neural networks can be used for recovering of any signal like video or wavelength.

Model: demosaic	
Layer (type)	Output Shape and # Parameters
input (InputLayer)	[(None, 128, 128, 4)] 0
conv2d (Conv2D)	(None, 64, 64, 16) 272
max_pooling2d	(None, 32, 32, 16) 0
conv2d	(None, 32, 32, 16) 2320
conv2d	(None, 16, 16, 16) 2320
max_pooling2d	(None, 8, 8, 16) 0
conv2d	(None, 8, 8, 16) 2320
conv2d_transpose	(None, 16, 16, 16) 2320
conv2d_transpose	(None, 32, 32, 16) 2320
add	(None, 32, 32, 16) 0
conv2d_transpose	p (None, 64, 64, 16) 2320
add	(None, 64, 64, 16) 0
conv2d_transpose	p (None, 128, 128, 4) 580
add	(None, 128, 128, 4) 0
conv2d_transpose	(None, 256, 256, 3) 111
conv2d_transpose	(None, 256, 256, 3) 84
Trainable parameters: 14967	

Figura 3: Demosaic network model.

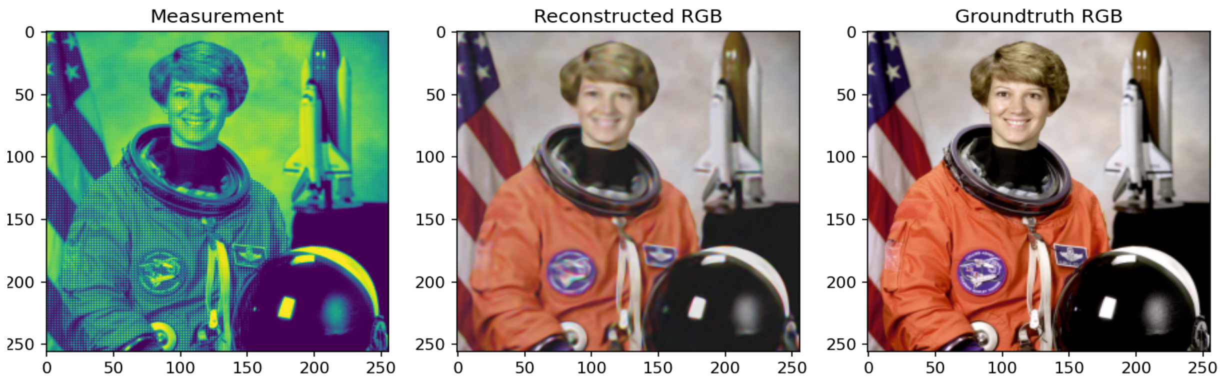

Figura 4: Demosaic reconstruction using implicit neural representation.

3. Exersices

- To download an app for your smartphone that allow you to recover the raw mosaic image and the RGB.
- Your task is using implicit representation and convolutional neural network as demosaicking algorithms. You have to recover the RGB image from the mosaic, see Figure 4. Consider the demosiacking neural network described in figure 3. Train the network as described in the text. Try to find an app for your smart phone that allows recording of raw format images. If you are unable to do this, find some raw format images online. Try to demosaic these images using the neural network, compare your results with the phone's built in demosaicing algorithm. Try improving the network of figure 3 by adding more layers or more convolutions per layer, train the network and compare results. Compare the neural demosaicing algorithm with demosiacing built into openCV, as in the text or with other demosaicing algorithms. The main point of this exercise is to gain experience training and using neural networks, so be sure to try several different variations on the network and training strategy. Discuss your experience training the network in your report.
- There is also datasets of RGB and Raw image file format digital negative (DNG) RGB and DNG images